



Airport security system

IBM outlines a method to secure an airport using a combination of AI, blockchain and biometrics. <https://digit.in/jan20-85>



Tackling human trafficking

A blockchain based human ID system can help tackle the problem of human trafficking. <https://digit.in/jan20-86>



Biometric attendance systems are extremely common now

centralised locations or on the cloud, this can cause them to flinch a little. To inject a healthy dose of security, transparency and trust into biometrics, blockchain technology can prove to be quite promising.

The current identity management model in most scenarios is entirely based on centralised cloud storage platforms. It paves a path for malicious threats by cybercriminals or hackers. If the cloud storage has vulnerabilities (and almost everything is vulnerable), hackers can manage to get hold of your sensitive identity data. The biometric data stored on centralised databases also have third-party vendors in the picture which further increases the threat to privacy. When it comes to online transactions, an identity management system usually consists of an end-user, a service provider, an identity provider and the control party or system. As stakeholders rise in number, so does the chances of data breach threats. Not only do stakeholders and cybercriminals pose a threat, but AI is also reaching a stage of development where it fool biometric systems. In November 2018, researchers from New York University found

a way to produce fake synthetic fingerprints using AI that could bypass biometric scanners. If hackers find a way into the centralised databases that store biometric data, they could employ AI technology to create synthetic fingerprints and wreak havoc on individuals' lives. Hence, there is a need for the biometrics

ID system to be injected with a dose of security and transparency which can be ushered in by blockchain.

ADVANTAGES OF INCORPORATING BLOCKCHAIN WITH BIOMETRICS ID SYSTEMS

Tapping the potential of blockchain can transform biometric ID systems by making them significantly more secure. Blockchain technology uses strong encryption algorithms to keep the data secure in decentralised ledgers that can ONLY be accessed by those by the one in authority, or if the authoritative figures have granted others access. There are no middlemen in this technology and it also has lower operational costs than centralised databases as well. By eliminating stakeholders or middlemen, traditionally complex transactions are made infinitely simpler between the sender and receiver, which saves time and the absence of third-party vendors makes it even more secure. Participants on the blockchain network can enjoy instant biometric authentication with low transactions costs, the lack of middlemen and overall, higher processing power. Combining these two technologies means that tech giants can develop seamless solutions for identity management and verification that keep the ID data safe in a distributed ledger system, thereby giving users complete control. With blockchain, user data is stored in distributed ledgers – meaning all the data from multiple individuals are not kept together. This means that there

is no singular point of failure in this system. Therefore, cybercriminals are less motivated to individually pick on identity data situated in numerous ledgers. These technologies which allow for self-sovereign identity management can be deployed by government agencies, financial institutions and enterprise businesses.

Additionally, our smartphones possess obnoxious amounts of biometric data stored in them. Online fraudsters are increasingly finding creative methods to gain unauthorised access to this data. Coupled with the 5,000 known security flaws in smartphones operating systems, and technologies such as Bluetooth which can also grant crimi-



Sirin Labs' blockchain-enabled smartphone

nals access to your phones, the risks are escalating day over day. Blockchain-based smartphones are already on the market and are being developed more as the technology becomes more mature. These smartphones can avoid attacks by cybercriminals to get access to biometric data since the data is stored in the blockchain itself. Blockchain is an immutable technology, therefore external, unauthorised manipulation is not possible which keeps our sensitive biometric data secure.

With blockchain-infused biometrics identity systems, the authorised individual has complete control of the decentralised, distributed ledger and can also actively view who accesses and shares the data within the ledger. The grasp of central powers and corporations over our identity data can be loosened and, in time, completely snatched away. 



Blockchain-based biometrics can alleviate the privacy threat